Run Sheet for `menuraR` Shiny App

1. Introduction

Thank you for participating in this study. We are testing the usability and effectiveness of the `menuraR` Shiny app, which allows users to compare multiple non-linear dimension reduction (NLDR) layouts and select the most reasonable one.

Your participation will help us improve the design and functionality of the app. This is not a test of your skills, but rather an evaluation of the app.

2. Tutorial

Please see this video for a quick walkthrough of the `menuraR` app. It only takes about 5 minutes to watch the video tutorial.

3. Getting Started

Follow the instructions below to complete the study. The task should take approximately 10 minutes

- Open the app in your browser: https://menurar.netlify.app
- Upload the High-dimensional data, Metadata, and Non-linear Dimension Reduction(s) provided for you.
- Here are some suggested layouts you can explore, but feel free to use the same or add any others you like:

Method	Hyper-parameters
tSNE	perplexity = 18
tSNE	perplexity = 7
tSNE	perplexity = 42

Method	Hyper-parameters
tSNE	perplexity = 26
UMAP	n_neighbors = 28, min_dist = 0.3
UMAP	n_neighbors = 45, min_dist = 0.55
UMAP	n_neighbors = 11, min_dist = 0.76

4. Tasks

Which of the layouts did you use for comparison?

Method	Hyper-parameters

What binwidth (a1) did you chose?

What is/are the best layout(s) provided by the app?

Would you choose the same layout(s) suggested by the app?

☐ Yes

☐ No

- **If yes**: Why do you think this is the best layout(s)?

____
____
____

- **If no**: What layout(s) would you suggest instead, and why do you think it would be better?

Layout(s): ____

Reason(s): ____
____
____
____

4. Demographic Information

Please answer the following:

1. Have you used principal component analysis in your work?

☐ Yes
☐ No

2. Have you applied non-linear dimensional reduction techniques such as tSNE and UMAP?

☐ Yes
☐ No

3. Select your subject area:

☐ Statistics
☐ Econometrics
☐ Bioinformatics
☐ Other: ___________

5. Completion

Thank you for your time and valuable feedback! Please remember to send the **Layouts plot image** and the **Layouts data** along with your responses.